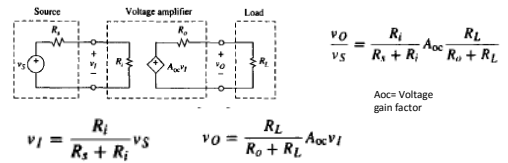


OPAMP Fundamentals

ARN



$$\frac{v_O}{v_S} = \frac{R_i}{R_i + R_i} A_{oc} \frac{R_L}{R_o + R_L}$$

Acc= Voltage gain factor

$$v_I = \frac{R_i}{R_i + R_i} v_S \quad v_O = \frac{R_L}{R_o + R_L} A_{oc} v_I$$

EXAMPLE 1.1. (a) An amplifier with $R_i = 100 \text{ k}\Omega$, $A_{oc} = 100 \text{ V/V}$, and $R_o = 1 \text{ }\Omega$ is driven by a source with $R_s = 25 \text{ k}\Omega$ and drives a load $R_L = 3 \text{ }\Omega$. Calculate the overall gain as well as the amount of input and output loading. (b) Repeat, but for a source with $R_s = 50 \text{ k}\Omega$ and a load $R_L = 4 \text{ }\Omega$. Compare.

Solution.

- (a) By Eq. (1.3), the overall gain is $v_O/v_S = [100/(25 + 100)] \times 100 \times 3/(1 + 3) = 0.80 \times 100 \times 0.75 = 60 \text{ V/V}$, which is less than 100 V/V because of loading. Input loading causes the source voltage to drop to 80% of its unloaded value; output loading introduces an additional drop to 75%.
- (b) By the same equation, $v_O/v_S = 0.67 \times 100 \times 0.80 = 53.3 \text{ V/V}$. We now have more loading at the input but less loading at the output. Moreover, the overall gain has changed from 60 V/V to 53.3 V/V.

FIGURE 1.3

(a) Op amp symbol and power-supply connections. (b) Equivalent circuit of a powered op amp. (The 741 op amp has typically $r_d = 2 \text{ M}\Omega$, $a = 200 \text{ V/mV}$, and $r_o = 75 \text{ }\Omega$.)

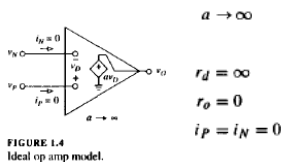
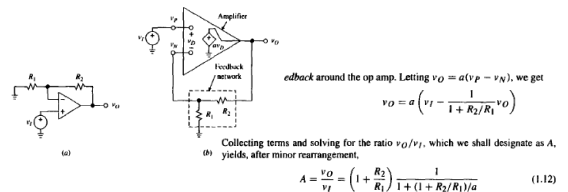


FIGURE 1.4
Ideal op amp model.



EXAMPLE 1.2. In the circuit of Fig. 1.6a, let $v_I = 1$ V, $R_1 = 2$ k Ω , and $R_2 = 18$ k Ω . Find v_O if (a) $a = 10^2$ V/V, (b) $a = 10^4$ V/V, (c) $a = 10^6$ V/V. Comment on your findings.

Solution. Equation (1.12) gives $v_O/1 = (1 + 18/2)/(1 + 10/a)$, or $v_O = 10/(1 + 10/a)$. So

(a) $v_O = 10/(1 + 10/10^2) = 9.991$ V,

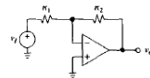
(b) $v_O = 9.990$ V,

(c) $v_O = 9.9999$ V.

The higher the gain a , the closer v_O is to 10.0 V.

Letting $a \rightarrow \infty$ in Eq. (1.12) yields a closed-loop gain that we refer to as ideal:

$$A_{\text{ideal}} = \lim_{a \rightarrow \infty} A = 1 + \frac{R_2}{R_1} \quad (1.1)$$



we have

$$v_P = 0 \quad (1.16)$$

Applying the superposition principle yields $v_N = [R_2/(R_1 + R_2)]v_I + [R_1/(R_1 + R_2)]v_O$, or

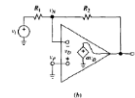
$$v_N = \frac{1}{1 + R_1/R_2}v_I + \frac{1}{1 + R_2/R_1}v_O \quad (1.17)$$

Letting $v_O = a(v_P - v_N)$ yields

$$v_O = a \left(-\frac{1}{1 + R_1/R_2}v_I - \frac{1}{1 + R_2/R_1}v_O \right) \quad (1.18)$$

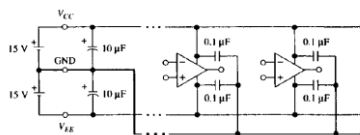
Comparing with Eq. (1.11), we observe that the resistive network still feeds the portion $1/(1 + R_2/R_1)$ of v_O back to the inverting input, thus providing the same amount of negative feedback. Solving for the ratio v_O/v_I and rearranging, we obtain

$$A = \frac{v_O}{v_I} = \left(-\frac{R_2}{R_1} \right) \frac{1}{1 + (1 + R_2/R_1)/a} \quad (1.19)$$



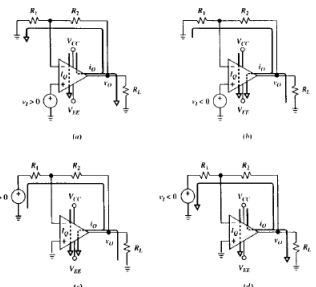
OPAMP powering

- 10uF polarized caps at power supply entry
- Ceramic 0.1uF capacitor at Vcc & Vee pins of opamp (mount in very close vicinity of opamp)



Powering Opamp

- ICC will always flow in VCC terminal
- IEE will always flow out of VEE terminal
- ICC = I_o + IEE @ VO+ or
- IEE = ICC + I_o @ VO-
- When I_o = 0 then ICC = IEE = I_Q: Supply quiescent current required to bias transistors in Opamp



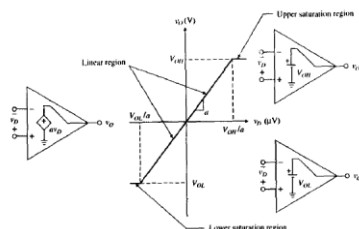
EXAMPLE 1.14. An inverting amplifier with $R_1 = 10$ k Ω , $R_2 = 20$ k Ω , and $v_I = 3$ V drives a 2-k Ω load. (a) Assuming $I_Q = 0.5$ mA, find i_{CC} , i_{EE} , and i_O . (b) Find the power dissipated inside the op amp.

Solution.

- (a) With reference to Fig. 1.37c, we have $v_O = -(20/10)3 = -6$ V. Denoting the currents through R_1 , R_2 , and R_L as i_1 , i_2 , and i_L , we have $i_1 = 6/2 = 3$ mA, and $i_2 = i_1 = 3/10 = 0.3$ mA. Thus, $i_O = i_2 + i_L = 0.3 + 3 = 3.3$ mA; $i_{CC} = I_Q = 0.5$ mA; $i_{EE} = i_{CC} + i_O = 0.5 + 3.3 = 3.8$ mA.
- (b) Whenever a current i experiences a voltage drop v , the corresponding power is $p = vi$. Thus, $p_{OA} = (V_{CC} - V_{EE})I_Q + (v_O - V_{EE})i_O = 30 \times 0.5 + [-6 - (-15)] \times 3.3 = 44.7$ mW.

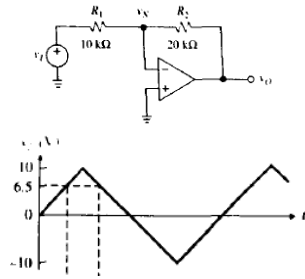
Output saturation

- Opamp o/p saturate below 2 volts of VCC and VEE
- For +15 volts VO = +13V,
- For +9 volts VO = +7V,
- Rail to rail opamps, CMOS based, with moderate loading VOH = VCC & VOL = VEE



??????

- Find waveform at V_o
- Find waveform at V_n



Soln.

